

Statistical Machine Learning - HW 5

Problems 1-4: Chapter 6

- Exercise #1, #3, #4, #10 from Chapter 6

Problems 5-6: Chapter 7

- Exercise #3, #6 from Chapter 7

Problem 7: Chapter 9 (SVM)

E.1 Soft-margin SVM

In real world applications, there are outliers in data. This can be dealt with using a soft margin, specified in a slightly different optimization problem as below (soft-margin SVM):

$$\min \frac{1}{2} \mathbf{w} \cdot \mathbf{w} + C \sum_{i=1}^N \xi_i \quad \text{where } \xi_i \geq 0$$

subject to

$$y^{(i)}(\mathbf{w}^T \mathbf{x}^{(i)} + b) \geq 1 - \xi_i.$$

ξ_i represents the slack for each data point i , which allows misclassification of datapoints in the event that the data is not linearly separable. SVM without the addition of slack terms is known as hard-margin SVM.

1. Intuitively, where does a data point lie relative to where the margin is when $\xi_i = 0$? Is this data point classified correctly?
2. Intuitively, where does a data point lie relative to where the margin is when $0 < \xi_i \leq 1$? Is this data point classified correctly?
3. Intuitively, where does a data point lie relative to where the margin is when $\xi_i > 1$? Is this data point classified correctly?

E.2 Kernel SVM

Support Vector Machines can be used to perform non-linear classification with a kernel trick. Recall the hard-margin SVM from class:

$$\min \frac{1}{2} \mathbf{w} \cdot \mathbf{w} \quad \text{s.t.} \quad y^{(i)}(\mathbf{w}^T \mathbf{x}^{(i)} + b) \geq 1.$$

The dual of this primal problem can be specified as a procedure to learn the following linear classifier:

$$f(\mathbf{x}) = \sum_{i=1}^N \alpha_i y_i (\mathbf{x}_i^T \mathbf{x}) + b.$$

Note that now we can replace $\mathbf{x}_i^T \mathbf{x}$ with a kernel $k(\mathbf{x}_i, \mathbf{x})$, and have a non-linear decision boundary.

In Figure 1, there are different SVMs with different shapes/patterns of decision boundaries. The training data is labeled as $y_i \in \{-1, 1\}$, represented as the shape of circles and squares respectively. Support vectors are drawn in solid circles.

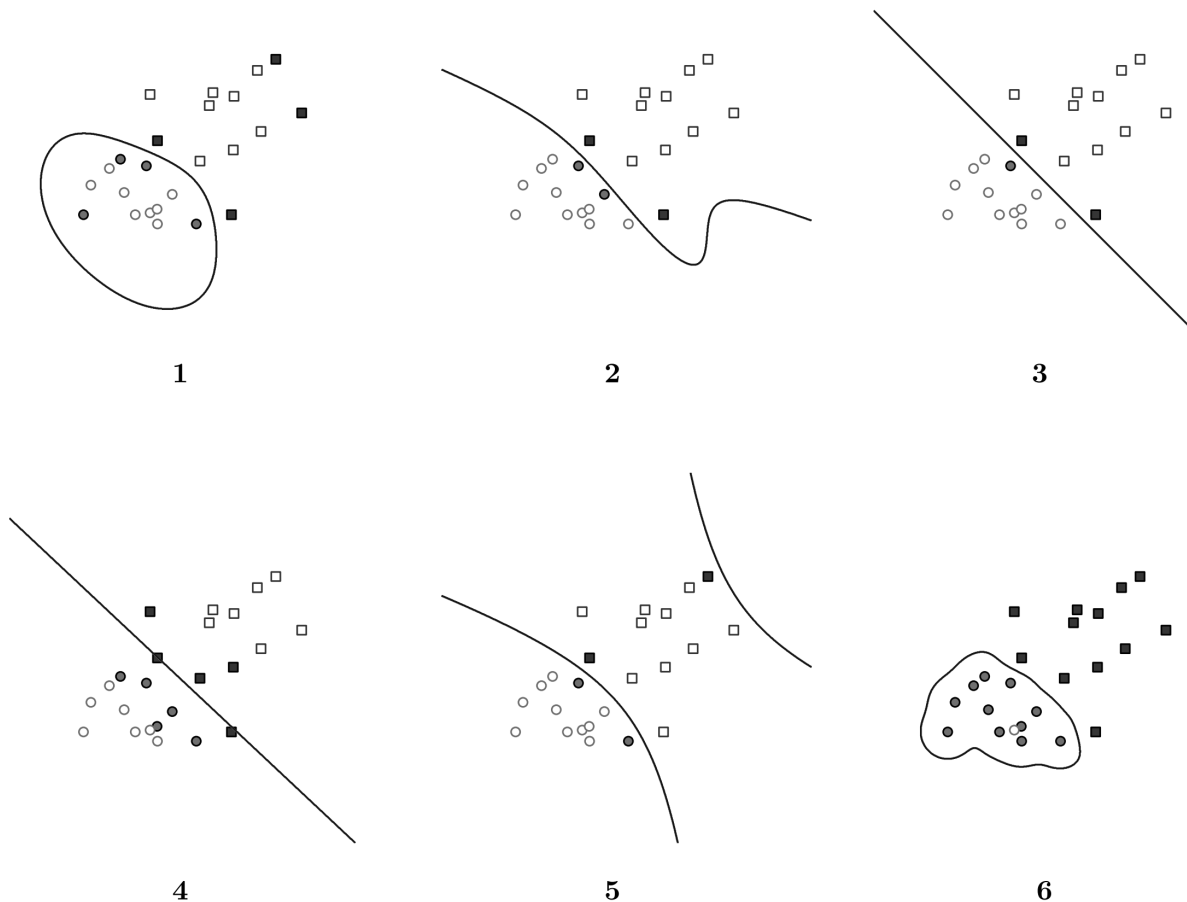


Figure 1: Figure 1: SVM boundaries

Match the scenarios described below to one of the 6 plots (note that one of the plots does not match to anything). Each scenario should be matched to a unique plot. Explain in less than two sentences why it is the case for each scenario.

1. A hard-margin linear
2. A soft-margin linear
3. A hard-margin kernel SVM with $k(u, v) = u \cdot v + (u \cdot v)^2$
4. A hard-margin kernel SVM with $k(u, v) = \exp(-5\|u - v\|^2)$
5. A hard-margin kernel SVM with $k(u, v) = \exp(-\frac{1}{5}\|u - v\|^2)$